

WASP-33b

R-0282

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-33_180205 · created 2026-07-17T15:48:51+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458154.6401 +/- 0.0094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.089 +/- 0.026
Transit depth (Rp/Rs)^2	0.8 +/- 0.46 [%]
Orbital Inclination (inc)	89.78 +/- 2.29
Airmass coefficient 1 (a1)	1.255 +/- 0.0064
Airmass coefficient 2 (a2)	-0.135 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	0.5 %
Best Comparison Star	None
Optimal Aperture	4.55
Optimal Annulus	10.67
Transit Duration (day)	0.1117 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.089 ± 0.026 vs 0.1066 (deviation 0.68σ)
Duration fit vs published	0.1117 d vs 0.1196 d (deviation 1.05σ)
Mid-time O–C	-1.5 min
Depth SNR	3.4
Residual scatter	0.5 %

Light curve

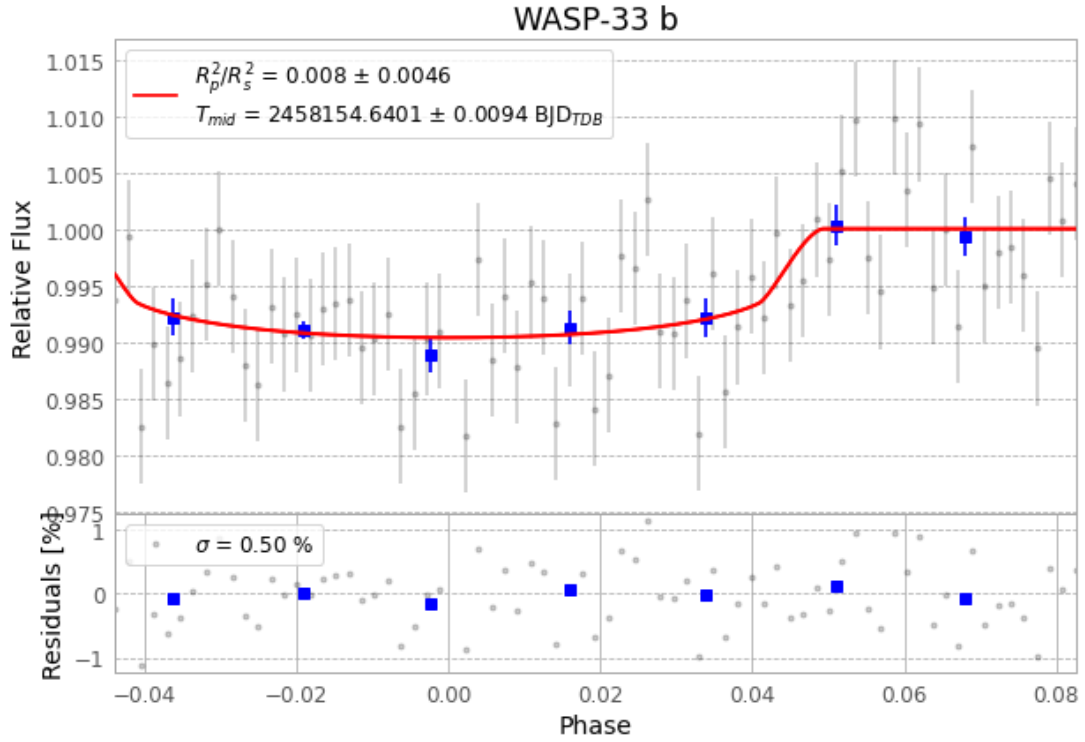


Figure 1 — FinalLightCurve_WASP-33 b_2018-02-04.png (SHA-256 f3d29cfd45042bc4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [288, 304], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b95987c48ceb83f3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

75 FITS frames (49 MB), WASP-33180205020312.FITS → WASP-33180205054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-33-180205.log (610ca9f01171...), FinalLightCurve_WASP-33 b_2018-02-04.png (f3d29cfd4504...), FinalLightCurve_WASP-33 b_2018-02-04.csv (557e08c8df00...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 15:48Z.